

Chapter I: Introduction

1 Preface

On May 11, 2021, Dr. Rochelle Walensky, Director of the US Centers for Disease Control and Prevention (C.D.C.), provided testimony to a hearing of the Senate HELP (Health, Education, Labor and Pensions) Committee on the subject of Covid and outdoor activities which proved after the fact to be surprisingly controversial. At issue was how Dr. Walensky defended the C.D.C.'s guidance on outdoor activities by indirectly citing a study which was a statistical outlier — later to be challenged by the same authors whose work she relied upon as a source of up-to-date information. The senators questioning Dr. Walensky focused on SARS-CoV-2 outdoor transmissibility, particularly in the context of when it was “safe” to reopen schools and summer camps. Outside of the committee itself, however, commentators attacked the C.D.C. not only for confusing guidelines, but also for how the C.D.C. director (so it was claimed) misleadingly cited the results of a recent scientific paper — even though (on closer examination) transcripts reveal that her comments were in line with the paper's actual text.

In retrospect, Dr. Walensky's statements did appear to give improper weight to one of multiple studies considered as part of a Systematic Review.¹ Hours after her testimony, one of the Systematic Review's authors used twitter to dispute the C.D.C. director's interpretation of their results. The specific issue in contention was an estimation of the percentage of Covid-19 cases that can be traced to outdoor rather than indoor transmission. This actual number is likely to be below 1%.² Dr. Walensky, however, cited the Systematic Review paper as claiming a transmission rate of *less than 10%*, a maximum bound derived from the statistical extremum amongst all the articles reviewed therein instead of a distributive average, a calculation not even attempted in the literature review paper itself.

It is worth noting that in the Systematic Review's very first sentence the authors report the same 10% upper bound, focusing readers' attention on that specific number, however much the study's text later paints a more granular statistical picture. In an interview with the New York Times, the author who criticized Dr. Walensky's testimony nonetheless argued that she and her co-authors “were very clear we were not making a summary number” with the 10% upper bound, and that their paper was technically a Systematic Review and not (as Dr. Walensky described it) a Meta-Analysis. From a scientific point of reference, as the New York Times correctly stated, “a meta-analysis often includes a precise estimate ... based on the data [whereas] a systematic review is more general.”

In short, the author implied that Dr. Walensky was misreading the analytic methods of their paper and as a consequence had presented a quantitative summary which was significantly different from their actual findings. Her 10% estimate was later cir-

culated in the media, creating an impression that outdoor transmissibility proportion could be close to such an upper bound, although the relevant statistical distribution in scientific reports actually skews much lower.

Of all the points of contention surrounding Covid-19, this particular controversy stands as little more than a footnote in the annals of US Government response to the pandemic (although confusing statements about outdoor transmissions continue to be cited as a contributing factor in the public's skepticism of the C.D.C.). However minor it may be, though, this episode raises interesting questions about how scientists and policymakers utilize scientific research. After all, we certainly believe that policies (especially in contexts related to medicine and health care) should be informed by empirical data. Yet, how should policymakers interpret research which actually produces empirical data sets? To the extent that many areas of large-scale public concern (certainly Covid-19 is a prime example) engender a number of different research projects, often yielding inconsistent results, how should such diverse data be consolidated into a single model for data-driven policy? Or, is there some threshold of divergence beyond which government officials should simply acknowledge that the science is inconclusive, and defer to scientists to produce more consistent findings before attempting to legitimize policies on empirical terms?

Insofar as multiple research projects often yield conflicting results, citing one specific research work can give a misleading overview of the relevant field as a whole — even if that work was conducted professionally and responsibly. In other words, our criteria for assessing the merits of scientific work inevitably shifts based on the relation of a given research endeavor to others on similar topics. Individual scientists no doubt can do no more than conduct their own research according to disciplined protocols, yielding results which are as conclusive as possible. In this sense well-executed research can be deemed definitive in the specific manner that it was conducted responsibly and in accord with protocols designed to minimize randomness and error. Findings which are *methodologically* sound may still be inaccurate. Whenever multiple research projects address similar themes, to the degree that their findings can be contrasted, the totality of results among multiple studies should presumably be considered as a context for assessing the claims of any one study individually.

These points may seem obvious, even trivial, but they raise interesting questions under the general rubric of a philosophy of science — research methods are supposed to adhere to rigorous protocols *so that* the corresponding work is empirically persuasive. Empirical science is driven by the belief that we can arrive at factually decisive results by conducting research which is optimized to yield statistically significant results. It is therefore at least philosophically uncomfortable when multiple studies — all done correctly and professionally — yield substantially incommensurate conclusions. Unless specific details in trial design or sample populations (whatever these may be in the research context) are identified which could explain scientific discrepancies, how can we be confident in the accuracy of individual (even well-implemented) scientific projects when his-

¹ See <https://www.nytimes.com/2021/05/26/briefing/CDC-outdoor-covid-risks-guidelines.html> for an overview.

² See the specific publication, “Outdoor Transmission of SARS-CoV-2 and Other Respiratory Viruses: A Systematic Review,” for details (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7798940>).

tory shows us many cases of equally meritorious research work yielding inconsistent results?

In principle, aggregative compilations of related research work — whether these are called “systematic reviews,” “meta-analyses,” or something else — can try to resolve the discrepancies between multiple related publications. Philosophically, however, we should consider whether this simply translates the epistemological uncertainties engendered by contradictory findings onto a different scale. After all, if multiple (well-executed) papers yield divergent results why should we expect a simply numeric average across all such papers to be more empirically accurate if we have no explanatory mechanism for what caused the discrepancies in the first place?

In the case of Dr. Walensky’s testimony, the manner in which she drew conclusions from the Systematic Review inadvertently placed undue weight on one or two specific papers included within that review — according to the New York Times, the 10% upper bound on outdoor versus indoor Covid-19 transmission may have been skewed by one Singaporean study among construction workers, whose working conditions are not representative of outdoor activities by everyday people in general. This demonstrates an indirect version of the fallacy in citing one single paper as a conclusive source when its findings conflict with other similar papers. And yet, data-driven policy has to draw facts from *somewhere* to be empirically grounded in the first place. It is not methodologically evident that some sort of quantitative synthesis of multiple papers should be deemed authoritatively more accurate than the results of one single paper. Is not meta-analysis itself one form of research, which can coexist with other meta-analyses potentially yielding different results in turn, and so on, *ad infinitum*?

Perhaps the best way to get beyond such an impasse is to be more rigorous in how multiple research projects are integrated. To the degree that similar research work yields a spectrum of discrepant results, scientists should actually try to explain those differences, rather than relying on some sort of statistical averaging effect. Thus, if findings from one project prove difficult to reproduce, scientists should try to identify the source of this difficulty in either the original work or the attempted replications, or some combination thereof. If one (well-executed) paper appears to be a statistical outlier, scientists should attempt to explain its divergent results.

Of course, some of this interpretive work doubtless occurs, informally and colloquially, as scientists conduct literature reviews or discuss the work of others amongst themselves. However, one could argue that more granular research integration and meta-analysis should be promoted as a formal mechanism within publishing platforms and scientific technology. On this theory, publishers should curate the tools allowing meta-analysis to be performed (more rigorously than today) and the software used to generate or analyze scientific data-sets should more aggressively implement procedures to integrate data spanning multiple research sources.

In recent years, the scientific community has indeed increasingly embraced open-access paradigms such as FAIR (Find-

able, Accessible, Interoperable, Reusable) and the Research Object protocol, which we will cite more explicitly in Chapter 2 and elsewhere. These initiatives are intended to make scientific data more transparent and open-access, promoting study replication and/or multi-project syntheses. To be sure, these same paradigms can also apply to meta-analyses themselves. Even when merely aggregating prior research work, there are methodological options which can affect the outcome of a meta-analysis, and the tactics and limitations of aggregative efforts can potentially be modeled systematically. Antecedent research in a meta-analysis (or systematic review) does not necessarily neatly align; some interpretive effort and data-marshaling may be needed to finesse disparate publications and data sets. A study of Covid-19 transmissibility *among construction workers in Singapore*, for example, is not precisely comparable to studies of outdoor SARS-CoV-D infectiousness in general. In order to have sufficient data in the first place, it may be necessary to rely on research which has certain reciprocal methodological anomalies along these lines.³ Nevertheless, in each case researchers can search for a formal mechanism to annotate and systematize the complications which arise from unifying multiple not-fully-compatible research environments into one integrated picture.

2 Data Integration, Hypergraphs, and Type Theory

There is, of course, no magic wand which anyone could wave over the worlds of science and publishing to re-engineer meta-analytic paradigms in a single historical inflection-point. Paradigm-shifts (notwithstanding Kuhnian narratives) tend to happen gradually, and in a decentralized manner. Anyone writing a single chapter or a single book can only hope to predict potential paradigm-shifts or articulate the disciplinary dynamics which could motivate them. Indeed, in this book we are trying to take such an evaluative perspective on the dynamic flow of research applied to the subject of biomedical data acquisition and data-integration practices, in both research and clinical settings.

Plenty of evidence suggests that scientists are starting to take transparent data sharing and research-replication seriously. We will consider the idea of “replication crises” in greater detail in Chapter 4; at present, though, we postulate that science has been rocked by a recurring failure to reproduce research results in follow-up studies, even when the original research appears to be well-executed. Separate and apart from the merits of a single research project, a string of similar projects which yield convergent findings have greater scientific weight — so that researchers have an incentive to spur replication studies to solidify the work they themselves publish. It logically follows that an emergent criterion for newly published research is *how well it facilitates* potential reproductions.

This situation does indeed portend a paradigm-shift which has ramifications in numerous scientific and technological domains. The value of having research consolidated through replication can in many contexts outweigh the value of keeping some data or methodological details private — for example, to protect In-

³As one example, the authors of the controversy-inspiring Systematic Review described in their paper having to exclude many papers from their analysis to begin with, which perhaps compelled them to retain the anomalous “construction workers” source, with an eye toward keeping a sufficiently large data set.

tellectual Property — thereby helping to expand the scope of the open-access data publishing ecosystem. Many publishers have likewise embraced an open-access publishing model, which is also an example of paradigm-shifts that have pushed the scientific community to embrace openness and transparency in sharing research data and methods. Scientists have a reputational stake in exposing their protocols to communal assessment, similar to how computer programmers profit more from open-access code than from closed-source alternatives.⁴

Against this background we can, therefore, see a certain narrative emerging in how contemporary scientific research is assessed: researchers have an interest in producing work which is (to the degree possible) easily replicable, transparent about methods and open vis-à-vis data access, and poised to be integrated with other projects on similar topics. Sometimes these paradigms are explicitly enforced/formally recommended by publishers or funding sources. The Bill and Melinda Gates foundation, for example, in their “guidelines for authors” (published as one specification within the FAIRSharing Initiative) essentially requires depositing open-access data sets on popular hosting platforms such as Dryad, Open Science Foundation, or GitHub as a precondition for fieldwork grants.⁵ Similarly, academic publishers strongly encourage authors to submit data sets to open-access platforms and to reference those platforms via “supplemental materials” and/or “data availability” sections of their articles.⁶ In particular, the onus is on authors to justify a *failure* to publicly share research data, such that pairing academic papers with open-access data is intended to be the norm rather than the exception.

Aside from merely observing the industry trends toward freely-shared research data and open-access research publications, we can also draw lessons from studying *why* these trends are becoming entrenched. What do scientists actually hope to accomplish by making research data publicly available? One facet of this trend involves double-checking research (and also statistical/computational) methods. The text of an article may condense or summarize findings into a single table or illustration, even just a single number (say, a 1% estimate for the proportion of Covid-19 transmissions which occur outdoors). However, researchers may find that these summaries are more convincing when they can demonstrate their provenance — perhaps by publishing data sets in their entirety rather than just statistical overviews, as well as sharing computer code (through which summarial calculations are attained), data-acquisition protocol descriptions, and so forth. In sum, such transparency in data sharing, among other benefits, can help prevent (or at least retroactively clarify) misinterpretations such as Dr. Walensky’s testimonial statements of outdoor Covid-19 infectiousness.

Aside from building trust in their research, data-sharing helps research projects prepare for potential replications, as intimated earlier, as well as for integration with comparable projects. These factors also augment the value of any one individual

project.

In short, among the factors driving current data-curation and publishing paradigms is the hope that new research will be attentively reviewed, synthesized with other research, and potentially reproduced/replicated in whole or in part, all of which can make new research more valuable to the relevant scientific community that is in a position to judge the work. In particular, *replication* and *integration* are key goals of new research: the potential for replication (by follow-up studies) and integration (with prior or future projects) are among the criteria by which new research is evaluated.

To the degree that these are indeed compelling motivations — researchers strive to model their text, data, and protocols to promote replication and integration — we can anticipate that research methods and tools (including software and computational tools) that amplify these aspects of research’s dissemination will be preferred. This can plausibly be deemed a driving factor in how scientific software, data modeling paradigms, and publication technologies will evolve in the immediate future: technologies will come to the fore to the degree that they promote research data integration and replication.

For a concrete example of these issues, consider clinical data-sharing networks such as OMOP (Observational Medical Outcomes Partnership), PCOR (Patient-Centered Outcomes Research Network) or CDISC (the Clinical Data Interchange Standards Consortium). These initiatives promote data-integration largely through conventional relational-database mergers, relying on Controlled Vocabularies or table-schema to enforce data field names and table columns. Conversely, Semantic-Web based data-integration projects such as the OBO (Open Biological and Biomedical Ontology) Foundry pursue data-integration via more free-form graph structures, regulated by Ontologies or “shape constraints” rather than by static table layouts (we discuss such indirect constraint logics in Chapter 6). A variation on graph-based integration techniques, based on property-graphs rather than Semantic Web networks, is evident in the University of Pennsylvania “Carnival” project, which we summarize in Chapter 3. Other model-sharing networks emphasize data models prioritizing computational simulations and Object-Oriented representations (we will cite specific examples in Chapter 4). Moreover, individual disciplines within biomedicine and bioinformatics have their own data-sharing protocols, often based on special-purpose file types and domain-specific software, several of which are reviewed in Chapter 2.

In short, researchers working in a biomedical context have multiple options for preparing publicly-shared data in preparation for potential integration with other studies (or with replication efforts seeking to reproduce their own specific work). Should data be modeled as relational tables, Semantic Web style graphs, property graphs, “objects” backed by Object-Oriented computer code, or as instances of domain-specific data formats? All of these data-representations coexist in contemporary technology with more or less comparable equilibrium, in the sense that no one paradigm dominates the others. How will data-representation strategies evolve in the future?

⁴Open-source projects get vetted and refined by a larger community of users and fellow developers; commercially licensed versions of open-source projects therefore have a trust factor which closed-source projects frequently lack, making open-source code often more competitive on the open market.

⁵See <https://gatesopenresearch.org/for-authors/data-guidelines>.

⁶See <https://www.elsevier.com/authors/tools-and-resources/research-data/data-guidelines>, for example.

Insofar as paradigm-shifts in science and publishing are being driven, as we claim, by priorities of data-integration and research-replication, we can anticipate that technical details such as data meta-models will evolve under pressures from these scientific considerations. Data representations which are conducive to replication and integration will likely become more widely used; less flexible models, such as relational-database technologies, may become deemphasized.

As large-scale initiatives such as OMOP, CDISC, and PCOR evince, the conventional Relational Database model remains influential. Nonetheless, over the past two generations the Semantic Web — and graph database technology in general — was predicted to substantially displace SQL-style technologies. That has not actually happened (even if Semantic Web formats such as RDF have indeed been widely used). One explanation for why predictions centered on the Semantic Web have fallen flat is that Semantic Web models, intended to be flexible and conceptually realistic, arguably fail by their own standards — this has engendered a trickle of computer scientists criticizing Semantic Web implementations more than motives, and presenting alternative models (such as Conceptual Spaces) which we will analyze further in chapters 6 and 9. Meanwhile, graph database technology itself is more general than the Semantic Web alone, and non-RDF formats (more expressive or structurally detailed than labeled graphs as such) have emerged as popular NoSQL database models, including hypergraphs and property-graphs. These technologies, too, may become increasingly influential in structuring how data is modeled for public dissemination and publishing in the future.

In short, we can predict that paradigms such as Conceptual Spaces, hypergraphs, and property-graphs will potentially become more substantial foundations for future data-sharing protocols which deviate from both Relational-Database and Semantic Web precedents. Work which synthesizes several of these developments — such as hypergraphs and Conceptual Space theory — is therefore especially interesting. One version of a unified hypergraph/Conceptual Space model has emerged in the context of Quantum Natural Language Processing, and our evaluation of that model’s assumptions, potential, and limitations will be an important focus of Chapters 6 and 9.

Supplemental materials for this book will be deposited on the Open Science Foundation, Dataverse, and GitHub.⁷ These materials include sample data sets which are encoded and annotated using techniques we discuss in Chapters 6-9. Such data sets are not intended for research purposes themselves, but rather to illustrate ideas about how data sets may be structured, to the degree that scientists really do prioritize expressive data models, microcitations (see below), integration with machine-readable text, and other data-publishing features mentioned in this introduction and elsewhere in the book. The supplemental repository also contains machine-readable text of the book’s individual chapters, encoded as separate documents using a special text-representation system designed to facilitate cross-references between publications and data sets. Finally, we include code li-

braries targeted at the sample data sets, in keeping with design patterns (discussed in later chapters) which assume that data sets will in general be published alongside code libraries providing functionality to read and manipulate the associated data, including parsers for the data’s serialization format.

By sharing code alongside data — optimized as necessary for the specific information comprising a data set — programmers can shift the burden of annotation and documentation to the computer code rather than the data itself. As we will discuss in Chapter 6 and elsewhere, source code presents a richer foundation (as compared to static data models) for pre- and post-condition annotations, requirements engineering, and other technical details which can express scientific theories and research protocols. Equipping data sets with custom-designed code libraries allows the data types uniquely instantiated via those libraries’ implementations to serve as models and documentations for the data set’s specific profiles; such a type-theoretic foundation, in particular, allows us to examine data set organization in a systematic fashion. We will consider type theory as a data-modeling paradigm in Chapters 5 and 6.

3 Philosophy and the Semantic Web

Data-exchange formats might seem like a mundane scientific topic, part of the minutia of research practice which academics attend to as a matter of professional competence, like curating bibliographic references, but hardly interesting outside its backstage role. It is curious, therefore, to consider that the Semantic Web has a colorful philosophical backstory and has had a relatively center-stage position in debates over AI’s potentials and limits — over whether human intelligence is mechanical enough to be digitally simulated.

The concept of Semantic Web *Ontologies* has become rather conventionalized, such that so-called Ontologies tend to serve essentially as Controlled Vocabularies or Taxonomies, constraining the classifications of data within structured information spaces, and the labels used to connect disparate data points. Ontologies are formally rooted in graph database technologies (or information spaces which emulate them, such as the Semantic Web itself). In this context, the principal elements of Ontologies are enumerations of labels which can be attached to graph nodes (providing a classification of the data set embodied by a graph) and graph edges (classifying the kinds of interrelationships that may be asserted between nodes). Ontologies are *controlled vocabularies* in the sense that conformant graphs may only utilize node or edge labels proscribed by Ontologies applied to the graph. They are *taxonomies* in that labeled terms may be sorted *hierarchically*: labels can name classification elements which are super- or subkinds, relative to other elements in their respective Ontology.

Philosophers understand the term “ontology” to name something of much greater metaphysical weight than just taxonomies on graph data-stores. Notwithstanding that background, the term “Ontology” was not chosen by accident; every *domain-specific* Ontology, essentially a data-model applied to empirical data sets, is understood by Semantic Web practitioners to be potentially unified into more general “upper” Ontologies, which

⁷For archive locations please visit <https://github.com/sciscape/DataIntegration-ConceptualSpaceModeling>.

have broader (and more philosophical) scope. Upper Ontologies aspire to a global classification of “objects” in general, both abstract and concrete, so that in addition to domain-specific concepts and definitions (*carcinoma is a kind of cancer*, say) one has broader metaphysical annotations (cancers are “disease processes,” tissues are “spatially extended regions,” and so forth). The rationale for this metaphysical superstructure is rooted in AI — Ontologies seek to endow scientific and technical data (particularly biomedical data, where Ontologies are especially popular) with a conceptual scaffolding analogous to human intelligence, insofar as we instinctively conceptualize objects through the lens of spatiotemporal extension, stasis and change, events and processes, and so forth.

Artificial Intelligence is, in fact, a key component of Semantic Web architecture, though in practice the role of AI is less on this metaphysical scale, focusing instead on the use of *axioms* and other Ontology constructions adding structural detail to Semantic data models. Ontologies employ axioms and annotations to assert constraints or patterns on how classifications and relations are used. For instance, Ontologies may assert that two relations are inverses (parenthood and childhood, say), or that one relation conceptually implies another — for instance, the relation of two people being *divorced* necessitates that they were previously *married*, such that at some prior point in time the *marriage* relation held. Instead of just sets of graph nodes and edges, then, Ontologies add logical structure to graph-form data: the *divorce* relation, for example, demonstrates how relations cluster into logical networks. Any database which represents a divorce-instance, and which is not fundamentally lacking data, would be expected to provide data about the necessarily antecedent *marriage*. We will return to *divorce* as a case-study in “multi-part” relationships in Chapter 6.

Ontologies, in short, are not just taxonomies or controlled vocabularies for graph databases; they also introduce axioms and logical constraints on graph-structured data. These extra constructions provide graph-based data models with added expressivity and precision. They also allow graph data structures to be targets for “reasoning engines” and other AI-driven analytics. Reasoning over the Semantic Web is analogous to query-evaluation, but uses methods rooted in Symbolic AI rather than the query-engine architectures typical of relational (or even NoSQL) databases. This is one sense in which the Semantic Web as a whole represents a “project” or even ideology closely aligned with AI itself. And the association between AI and the Semantic Web has also been a source of criticism, either from perspectives which are skeptical about the more holistic claims of AI (or “Artificial General Intelligence”) visionaries, or those which feel that the Semantic Web’s fairly modest data-representation paradigms do not fully harness the power of AI (or some combination of the two).

This is some of the milieu in which technical-sounding debates as to optimal data-representation meta-models become invested with unlikely philosophical gravitas. Some of the influential figures in Semantic Web evolution (and criticism) come from the realm of philosophy and humanities/linguistics, not from science (or computer science), such as Barry Smith — a scholar whose

earlier work was grounded in Phenomenology and Central European Philosophy, and who pivoted mid-career to information science, spearheading the OBO Foundry — and Peter Gärdenfors, a linguist who originated Conceptual Space theory and has catalyzed certain counter-narratives critiquing the Semantic Web (mentioned earlier).

At some level these are relatively minor episodes in Intellectual History, and of course the philosophical germination of the Semantic Web has relatively little bearing on a researcher who adopts a specific OBO Ontology, for example, as a structuring device for their data. But philosophical controversies around the Semantic Web help alert us to what is at stake in data-sharing protocols. Why is a vision such as the Semantic Web — a globally synthesized network of knowledge subject to common meta-models which can be concretized in domain-specific standards, potentially synthesizing data from myriad sources — popular? What lies behind its intuitive appeal? The underlying dynamic in debates about (say) Conceptual Spaces versus Ontologies appears to be rooted in scientists’ search for data *representations* which seem to intuitively capture the theoretical commitments and conceptual architecture surrounding scientific data, its research origins, and its technical import. In short, scientists seem to consider research data not only as a digital artifact to be shared, but as an embodiment of a given scientific perspective and research environment. Data sets, on this point of view, should *communicate* something about the science and research that produce them, as well as serving the practical goals of moving data from one point to another.

Implicitly, then, in the following chapters when we talk about “data sets” we will usually be considering collections of information which are more than just “raw data.” Data sets, specifically, are organized and documented to convey some details about the data set’s scientific origins. The data *models* which govern how data sets are encoded thereby need to do more than simply digitize raw data in an unambiguous fashion. Instead, data models have to permit data-set curators to use the organizing principles and annotation mechanisms within each data set as communicative tools representing the appropriate scientific and theoretical background. Such requirements point beyond the formats typically used for data encoding in the past (XML, JSON, HDF, ASDF, and so forth); we will examine criteria and architectures for more expansive and conceptually intuitive formats in later chapters.

4 Covid, Philosophy, and Science

A philosophical discussion which takes the Covid pandemic as a point of departure would be incomplete without acknowledging how some facets of associated public-health policy, such as mask or vaccination requirements, have become politicized. While Covid-19 serves as a catalyst and magnifier for some of these issues, underlying questions — about the reliability of scientific data, the proper balance between public health and individual rights, and so forth — surely transcend any single disease. It is worth pointing out several representative controversies where technical issues related to scientific data and publishing are directly relevant.

In the context of vaccinations, for example, consider the noteworthy case of a retracted article which (not quite appropriately) became widely cited by so-called “anti-vaxxers” — that is to say, embroiled in the political machinations of vaccine policy. The contested article (published on June 24, 2021, in the journal *Vaccines*) attempted to quantify the correlation between Covid-19 deaths prevented by vaccination against deaths which *followed* vaccination. The key detail appears to be conflicting interpretations as to whether the authors claim that vaccinations had *resulted in* mortality or merely (acausally) *preceded* fatal outcomes in some circumstances.

After several editors’ resignations (some later returned to the journal), *Vaccines* retracted the article, prompting in response a clarification from the authors which appeared to walk back some of their claims. According to *Science Magazine*, the authors conceded that “Currently we only have association ... we never said anything else.”⁸

At issue is *first* estimating the number of deaths *prevented* by vaccines (an imperfect process, because one cannot precisely quantify counter-factuals) and *second* distinguishing deaths *caused by* vaccines as opposed to fatal incidents which fit adverse-reaction criteria — so they were *potentially* vaccination side effects — but had other feasible medical explanations as well. Even with that caveat, the authors cited a very low count of fatal incidents (roughly 4 per 100,000) but also (via likely flawed methods) derived a small figure for the hypothetical number of fatal Covid-19 cases prevented by vaccines, on average. Combining both numbers led the authors to summarize that 2 lives are lost for every 3 saved by vaccination, an almost surely inaccurate calculation which was seized upon by anti-vaxxers, who circulated this figure outside its proper context and methodological nascence.

Competent scientists quickly pointed out the article’s misleading use of adverse-effect reports as well as their dubious derivation of vaccines’ efficacy (based on a single Israeli study comparing vaccinated and unvaccinated cohorts), but the controversy lived on precisely because scientists questioned how the paper survived peer review in its later-to-be-retracted form to begin with. Even if non-scientists with a political agenda selectively exploited certain details without due consideration for context, peer review did confer an imprimatur of merit on the original work which proved to be premature, but only after having already providing cover for anti-vaccination claims.

A second controversy which presents thought-provoking interpretive questions is the so-called “lab leak” theory, according to which SARS-CoV-2 escaped from a Wuhan research facility rather than having a zoonotic origin. The preponderance of scientific opinion appears to be that theories of the virus being manually engineered are *plausible*, but also that every epidemiological and genomic factor cited as potential evidence of non-zoonotic origins have alternative (and more mundane) explanations.⁹ A notable incident reflecting this dynamic involved Nobel laureate David Baltimore, who was quoted characterizing

certain RNA details as “smoking gun” evidence that SARS-CoV-2 was bioengineered. One scientist who led a team investigating possible “genome-tampering,” Kristian Andersen (of the Scripps Research institute), refuted Baltimore’s arguments by pointing out that the **CCG** codon accounts for about 3% of the SARS-CoV-2 nucleotides encoding arginine (an amino acid), which is the particular detail most often claimed to arise only via human manipulation (SARS-CoV-1 has an even higher rate of this specific encoding, even though little “bioweaponry” speculation accompanied that initial SARS outbreak). Baltimore in an email to *Nature* then moderated his earlier comments, claiming only that “there are other possibilities and they need careful consideration, which is all I meant to be saying.”

To some degree such give-and-take of claims and interpretations is a natural (and healthy) aspect of science, except that (most scientists would probably agree) conducting this exercise through public forums, in a politically charged environment, ends up obscuring the weight of scientific evidence. Insofar as those who wish to advance specific theories or policies (opposing proof-of-vaccine mandates, say, or trying to geopolitically leverage claims that the Chinese government — whether out of malice or incompetence — instigated the pandemic) can find backing for their positions from tendentious peer-reviewed writing and/or decorated experts, the possibility of scientific evidence strongly indicating certain interpretations over others becomes diluted.

At stake here is apparently, again, the contrast between individual studies and the “preponderance of evidence,” which in principle should constrain the impact of any single investigation, especially in areas where many research projects overlap and where a particular analysis contradicts the majority of juxtaposable findings. The general maxim that multiple studies yielding similar results carry proportionately greater weight is straightforward, but arguably it remains an open question how to formalize this intuition on a philosophical or meta-scientific level. Notions such as the “weight of evidence” aggregated across multiple studies, or metrics of findings either supporting or deviating from prior research, have not been codified at the meta-analytic scale with the degree of entrenchment that paradigms of “scientific method” are accepted (philosophically as well as operationally) for research projects singularly. One can anticipate that such meta-scientific questions will be explored more rigorously as the overall paradigm of open/transparent data sharing becomes further entrenched.

5 How Cancer, Covid, and Cardiac Care May Accelerate Emerging Research Trends

It has been suggested that the Covid-19 pandemic may have a lasting impact on many societies partly by accelerating trends which were already latent, such as allowing a greater proportion of office-style work to be performed remotely. In scientific terms, it is hard to overestimate the significance of rapid vaccine development: given that most post-infection interventions have only had limited success in serious cases, it was only via mass vaccinations that fortunate jurisdictions have been able (as of mid-2021) to resume some semblance of normalcy. Moreover, companies

⁸ See <https://www.sciencemag.org/news/2021/07/scientists-quit-journal-board-protesting-grossly-irresponsible-study-claiming-covid-19>.

⁹ See, e.g., <https://www.nature.com/articles/d41586-021-01529-3>.

were able to develop their vaccines quickly because of considerable research carried out, especially vis-à-vis RNA vaccines, well before 2019. The lesson many scientists and policy-makers may take away from this history is first that an infrastructure should be put in place for rapid vaccine testing, development, and production — anticipating future pandemics — and second that fundamental science relevant to vaccines can pay dividends in expected ways. In the case of SARS-CoV-2, the science of RNA vaccines become consolidated just before a disease would emerge for which that science would prove invaluable.¹⁰

One can speculate that the Covid-19 fallout will also leave a residual effect in areas such as scientific publishing, research data management, and the interface between science and public policy, for reasons already intimated. We have cited examples of civil administrators inciting unexpected controversy even with recommendations that are rooted in up-to-date science; of peer-reviewed (and at least superficially credible) research getting adopted by anti-vaxxers with an agenda that is *prima facie* fundamentally anti-scientific; and of a Nobel laureate essentially dialing back relatively informal (but vaguely sensationalist) comments that fed into the spy-novelsque intrigue of germ warfare.

Trying to reach conclusions from such disparate events may connote a fallacy of some homogenous “publishing industry” or “scientific community” or “academic establishment,” all of which are of course part of a diverse and decentralized social/commercial milieu; nevertheless, some cautious observations may be warranted.

One plausible point of argument is that the proliferation of scientific work *along with* the emergence of data-sharing and data-curation paradigms *jointly* imply that the meaning and notion of “peer review” is noticeably evolving. In this context it is worth referencing analyses such as those by Todd Carpenter (Executive Director of the National Information Standards Organization) on the very nature of peer review applied to *data* rather than *documents*.¹¹ As this (and similar) studies point out, the process for assessing published data sets — if we stay in the context of peer-review for conventional scientific *literature* — is much less standardized than for publications themselves.

In particular, the fact that an author transparently presents research data as supplemental materials to scientific writings may check one box in a text’s favor; however, reviewers might not then proceed (when vetting submissions in general) to closely examine the data sets themselves, with an eye toward measuring them against disciplinary norms. Some scientific groups have accordingly proposed “quality standards” based on factors such as ease of data/code reuse, thoroughness of meta-data, adherence to data-sharing protocols, and the relevance of data to its associated articles. Such standards may then influence how scientific work is received by government officials or the public writ large, because the scientific community could indicate whether publications have been developed in the context of high-quality (or, conversely, sub-par) data-curation, adjoining that assessment to other factors (such as peer review or journal reputation) which

individuals or policy-makers use to construe scientists’ merit in the eyes of other scientists.

Moreover, practically and logistically speaking, we should recognize that data-curation is time-consuming and, in its most rigorous forms, embodies disciplinary knowledge which stands apart from natural sciences themselves. To the degree that diligent data-curation expectations serve as one criterion of research merit, we can envision an evolution in how research teams and/or projects are set up, with proper time and resources allotted to data-management concerns alongside other priorities (those implicated in data-acquisition to begin with).

One could, specifically, envision a gradual conceptual shift away from reading research papers as “static” documents, insofar as data sets (and, where applicable, research code) may be able to cycle through multiple versions even after publications appear in final form. Version-control systems and branch/clone technologies can allow scientists to keep track of how data and/or code associated with a specific project is evolving while still maintaining the integrity of data sets as citable assets. In this context, scientists may increasingly perceive conventional publications as summaries rather than the substantial core of their research work, and place equal weight on Research Object style resources that can evolve more flexibly/dynamically and over which they have greater editorial control. Likewise, interactive data-sets — which can embed multi-media visualization capabilities beyond the scope of ordinary print documents (e.g., hooking up a publisher’s portal with radiology image data so that readers while reading an article or book could see in an engaging 3D format a radiographic image of cardiac tissue, an image of differentiated cancer cells, or an image of ground-glass opacity of lung tissue in Covid patients) — might become esteemed as pedagogic tools helping researchers to exposit their theories and methods (we revisit this point in greater detail in Chapter 4).

Although such a shift in priorities could potentially complicate the concept and process of peer review — because the “work” to be assessed becomes more of a moving target — acknowledging the evolutionary nature of research projects may help restore public confidence in assessments of scientific merit, because it would signal the relevant industries’ acknowledgement that appraisals of scientific work are inherently provisional and ongoing. Cases such as *Vaccines*’s retracted article actually reveal science appropriately correcting itself; however, such lingering controversies suggest that non-scientists entertain a misguided conception that research can be crisply sorted into bins of “merit” and “demerit,” a simplification which has no basis in science itself.

One may also anticipate the emergence of more rigorous meta-models for scientific data — employing constructions such as type theory or Object Models to classify the basic units of data sets insofar as these comprise resources with their own histories and quality-control standards. For example, “microcitations” are understood to link natural-language texts with *parts* of data sets, but this definition is inexact: what is a “part” in this context (one record, data-point, measurement, calculation, etc.)? In order to standardize data-curation assessments, scientists need to define basic formulations such as data sets’ version-history (e.g., what are the units of change as a data set gets updated)

¹⁰ See <https://www.statnews.com/2020/11/10/the-story-of-mrna-how-a-once-dismissed-idea-became-a-leading-technology-in-the-covid-vaccine-race>.

¹¹ See <https://scholarlykitchen.sspnet.org/2017/04/11/what-constitutes-peer-review-research-data>.

and overlap/reuse (e.g., what are the units of content which could adjudicate priority claims, plagiarism, impact factor, and so forth)? We explore type-theory in the data curation context (mostly in Chapters 5 and 6) partly because these structural questions seem poised to become increasingly important for the relationship between data-curation and the overarching scientific (and publishing/dissemination) process.

6 Navigating the Proliferation of Research Data

Whatever the motivations of scientists in curating research data, a further ineluctable detail of contemporary science and publishing is the large volume of (meritorious) work being produced. No doubt, it is better to have more good science than less, but the sheer scale of research work presents a challenge both to individual scientists (who are charged with making their contributions known to the relevant communities that can leverage them) and to technologies powering science as a whole.

Since it is impossible for any one person to read all scientific literature produced at any one point in time — or even all literature confined to a single specialty area, such as oncology, cardiac care, or Covid-19 — scientists envision AI tools that could guide researchers toward relevant papers and data sets. More advanced search engines for scientific documents — if these can actually be implemented — would help investigators cut through the mass of literature and hone in on specific studies which are precedents or theoretical foundations for their own work. In short, better search tools could counteract the challenges posed by rapid expansion in the volume of scientific work. Scientists might then have the best of both worlds: a vibrant scholarly community which produces large quantities of credible science and, simultaneously, technologies to obviate information overload.

This is an encouraging idea, but there seems to be little evidence that search tools specifically designed for science perform noticeably better than generic web searches. At best, truly accurate publication-search capabilities appear to remain a project for the future. A good case-study in the current state and limitations of publishing technology is offered by the CORD-19 corpus, curated by the Allen Institute of Artificial Intelligence, to promote text and data mining targeted at literature related to SARS-CoV-2 and (to the degree that they may benefit Covid-19 research) coronavirus studies in general. The CORD-19 compilation provides machine-readable full-text versions of over 280,000 publications (as of mid-2021). We review the features and architecture of CORD-19 in Chapter 3. For the moment we will simply point out that the data scientists who formulated CORD-19 openly acknowledge limitations in their methodology to obtain full-text representations and to curate them in a searchable manner. They even suggest the need for “a call to action” encouraging publishers to develop “distribution formats [for] scientific papers” which are less ambiguous than PDF (i.e., less opaque in text-encoding), to share publication texts in “structured format[s] like JSON, XML, or HTML,” and to embrace “strict schemas” for article meta-data.¹² If the availability of machine-readable

text serves as a qualification distinguishing which papers are included in aggregative corpora — and also which papers, once included, are more prominent in search results due to their being properly annotated (with demarcated keyphrases, findable data links, and so forth) — then publishers have an incentive to adopt paradigms akin to those which the Allen Institute is advocating in this context.

Similar challenges confront finding and integrating research data across disparate projects. It is difficult to search within data sets because there is no obvious foundation to look for key phrases, for instance, the way that search engines can match search terms against the raw text of a publication. There is, in general, no “raw text” within a data set that can be scanned for keywords and phrases. A related problem is that multiple data sets can be hard to aggregate together, even if they use similar methods applied to similar real-world problems. Unless there is a rigorous isomorphism between the statistical parameters and data-types employed between two kindred research projects, there is no automatic process to map one project’s parameters onto another’s so that they may be analyzed or visualized as a whole, or subjected to integrated statistical processing. Even subtle differences in parameters’ variance, distributions, ranges, and data-acquisition methods complicate attempts at data aggregation spanning two or more research projects.

In light of these difficulties, scientists have proposed numerous formats for describing published data sets, with the hope that common representations would make data sets more searchable and interoperable. One aspect of these standardization projects is the notion of “microcitations” (viz., strategies to demarcate individual parts of a dataset as citable references), analogous to citations of specific pages within a published document. The process of forming microcitation-targets in the context of specific data sets, however, depends on the format through which the data is encoded. Object values in JSON, SQL table rows or columns, XML document nodes, or record-tuples for formats such as **numpy** or CSV may all be feasible microcitation sites. Given that data sets might employ any of these representations (or many others) it is not unproblematic to standardize a general-purpose micro-citation format.

These data-sharing and text-mining challenges are significant, but they also give us a lens through which to anticipate what kinds of data curation and document-preparation technologies will become popular in the next phase of scientific publishing. It is reasonable to guess that scientists will benefit from standardized, multi-disciplinary data representations that support micro-citations, that allow publication texts to cite specific parts of data sets (much as they cite other articles), and that allow code to be re-used across multiple research projects as a means to achieve data integration. We consider this a hypothesis as to the general priorities that will shape scientific computing and publishing technologies moving forward. The actual software engineering and data-modeling structures and design patterns that might realize these general goals will be the subject of much of our analyses in several later chapters, particularly Chapters 5, 6, and 9, and (in the specific bioimaging context) Chapters 7 and 8 as well.

¹² See Lucy Lu Wang, *et al.*, “CORD-19: The COVID-19 Open Research Dataset” (<https://arxiv.org/pdf/2004.10706.pdf>), page 8.

7 Summary

Most of the chapters in this book will be focused on different approaches to data modeling, such as Type Theory, Conceptual Spaces, or Graph Database architectures. Our emphasis from a *theoretical* point of view will be on data-modeling representational paradigms whose goals and criteria are oriented toward software engineering and the interoperation of distinct software components. That is to say, we advocate for data-representations whose rules and conventions prioritize the implementation of software components which produce, share, and consume the modeled data. As an underlying assumption: any database, data set, or information space should be engineered with the expectation that multiple (not fully isomorphic) software components will be interacting with that data; and that parts thereof will be passed and shared between such components, implying that data should be structured to facilitate cross-component communication.

We propose that most of the theoretical constructions introduced vis-à-vis hypergraph or code models may be concretely instantiated through virtual machines via which query-evaluation engines may be implemented. A full exposition of the design and construction of such virtual machines is outside the scope of this book, but we present several analyses here that serve as precursors to a more formal elucidation of hypergraph-query virtual-machines implementations. More broadly, we use the architecture of virtual machines within this category as an organizing motif for analyses conducted in Chapters 6 and 9.

From a more practical or “applied” point of view, we will call attention in particular to biomedical research projects which synthesize information with variegated disciplinary provenance and diverse data profiles. Of course, much biomedical research is inherently interdisciplinary. However, new breakthroughs and new research methods and technologies have accelerated the cross-disciplinary insights of research in several specific biomedical disciplines, yielding diagnostic, prognostic, and explanatory models that cut across biophysical scales (molecular, cellular, tissues, organs) and data-acquisition modalities (proteomics, genomics, biopsies, image processing, lab assays — such as for biologic sample analysis — and so forth). Examining literature where these integrative studies are described, it becomes clear that scientists often construct the software ecosystem powering their research in ad-hoc ways, piecing together diverse software components (sometimes standalone applications, sometimes code libraries, or some combination of the two) designed for specific disciplinary contexts.

We present arguments in later chapters to the effect that the relatively informal trial-and-error approach often taken to integrating multi-disciplinary biomedical data can act as an impediment to research replication and the systematic evaluation of interdisciplinary research findings. This is one reason for engaging in a detailed review of data profiles, data modeling paradigms, and data integration techniques, so as to lay the foundation for a software ecosystem which can support the emerging paradigm of transparent and replicable research data and digital scientific resources.

We do not claim any special insights into interdisciplinary biomedical methods or data-sharing as such — it is commonly acknowledged that data sharing is an increasingly important part of both research and clinical practice, and that breakthroughs in fields such as oncology and immunotherapy will depend on carefully calibrated multi-disciplinary data integration. However, while undoubtedly (in light of today’s highly interconnected digital-health ecosystem) many biomedical and clinical data spaces are utilized by multiple (independent) software components, there are intricate design challenges which confront the engineering of data sources that allow autonomous components to leverage their data in consistent (but flexible) ways. Our biomedical software ecosystem, we argue, still remains more fragmented and unsystematically designed than would be warranted given how profoundly different biomedical subdisciplines have been interconnected in recent years.

In particular, we examine circumstances through which interoperability impediments contribute to antipatterns we refer to as “ecosystem fragmentation,” and (in Chapters 4, 7, and 8) describe what we call “multi-aspect modules” as potential corrective strategies. We use *multi-aspect* to qualify modular design patterns aiming for components intermediate in scale between domain-specific code libraries and monolithic scientific applications. Here the goal is to merge benefits of standalone applications (in particular, combining multiple software-engineering concerns, such as GUIs, data persistence, and data sharing/serialization, into a single code base) with those of smaller-scale code libraries (viz., interoperability, and the flexibility of mixing modules into standalone applications in different combinations, tailored to the needs of individual projects).

As for Dr. Rochelle Walensky’s testimony whose controversy was discussed earlier, the C.D.C. director might be justly held accountable for misrepresenting a specific Covid-19 study, but she can hardly be faulted for attempting to base her testimony on peer-reviewed scientific literature. The problem is that — although many people believe government policies should be grounded on scientific evidence and should respect scientific consensus wherever possible — all too often there simply *isn’t* scientific consensus, even in light of substantial real-world data. Such lack of consensus should not inhibit policymakers from basing government decision on data-driven, empirically-minded deliberation, but it implies that scientists need a more sophisticated model of how to translate scientific findings into public policy insofar as the science itself is sometimes inconclusive and contradictory.

One way to achieve this, as explicated in this volume, is by formulating more sophisticated presentations of research data, of archives tracking multiple research projects, and of document-preparation in conjunction with data curation. Equally important are the software and algorithms used to integrate disparate data sources (while also modeling the anomalies and structural anisomorphisms that can make integrations inexact) — ideally leveraging innovative ideas in query-engineering and data models that come to the fore within the bounds of a sufficiently multi-disciplinary perspective, some of which we hope to have advanced here.